GC University, Lahore

SUBJECTIVE QUESTION

EXAMINATION: Fall 2021
TITLE: GRAPH THEORY
TIME ALLOWED: 155 Minutes

COURSE CODE: CS- 4102
SEMESTER: 5th
MAX MARKS: 40

Note: Attempt any **FOUR** questions from the following.

Q#	Question	Marks

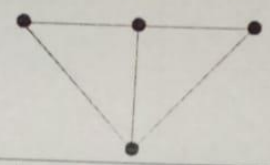
2 — Marks: 2+2+3+3

a) How many vertices are needed to construct a graph with 6 edges in which each vertex is of degree 2.

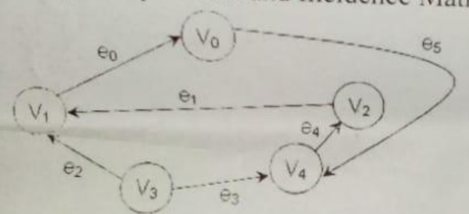
b) It is possible to draw a simple graph with 4 vertices and 7 edges? Justify.

c) A tree has five vertices of degree 2, three vertices of degree 3 and four vertices of degree 4. How many vertices of degree 1 does it have?

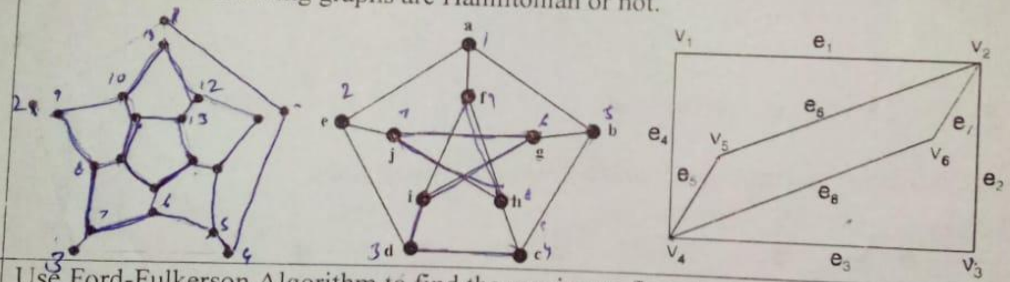
d) Find all spanning trees of the graph G given below:



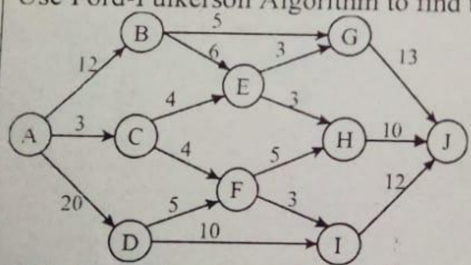
3(i) Find Adjacency Matrix and Incidence Matrix of the given directed graph. — 4



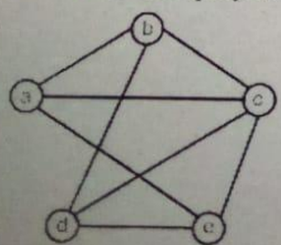
(ii) Show that the following graphs are Hamiltonian or not. — 6



4(i) Use Ford-Fulkerson Algorithm to find the maximum flow for the following network. — 5



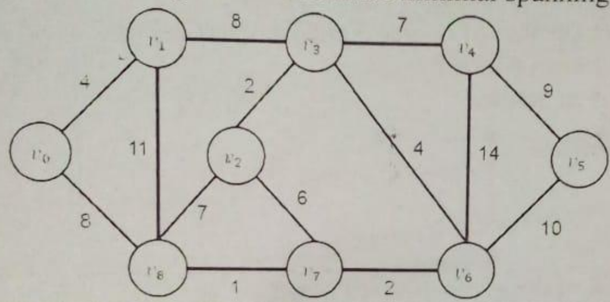
(ii) Find chromatic polynomial of the given graph. — 5



5(i) Weights are given for the edges between 7 vertices, labelled A to G. Find the minimal weight spanning tree. What is the total weight of this spanning tree? 5

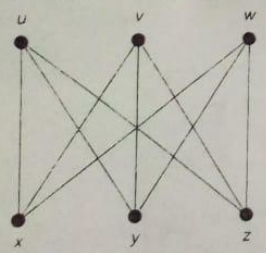
A						
11	B					
17	9	C				
17	12	14	D			
11	17	15	10	E		
16	9	9	10	8	F	
20	10	21	19	8	12	G

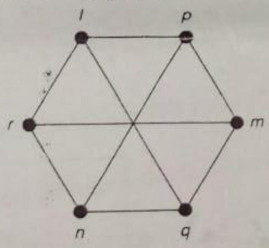
(ii) Use Prim's Algorithm to find the minimal spanning tree of the following graph. 5



6(i) Draw a graph that should be Hamiltonian but not Eulerian. Similarly draw another graph that should be Eulerian but not Hamiltonian. 4

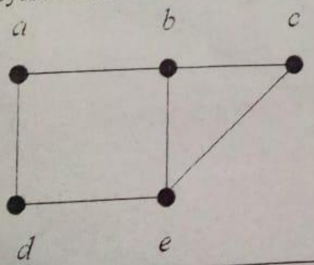
(ii) Show that the given graphs are isomorphic. 3



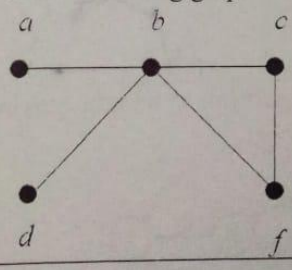


(iii) Find symmetric difference, intersection & union of the following graphs. 3

G_1



G_2



COURSE CODE: CS-4102

TITLE: Graph Theory

Time allowed: 155 minutes

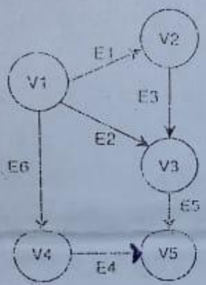
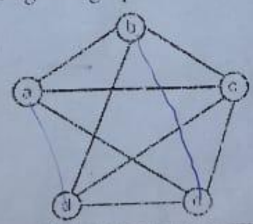
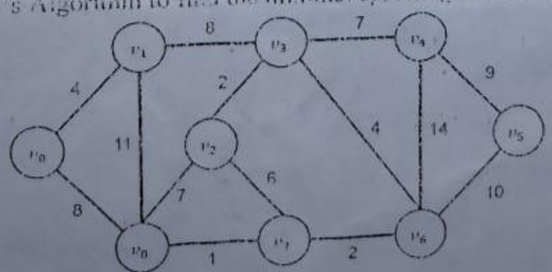
Marks: 40

Name:

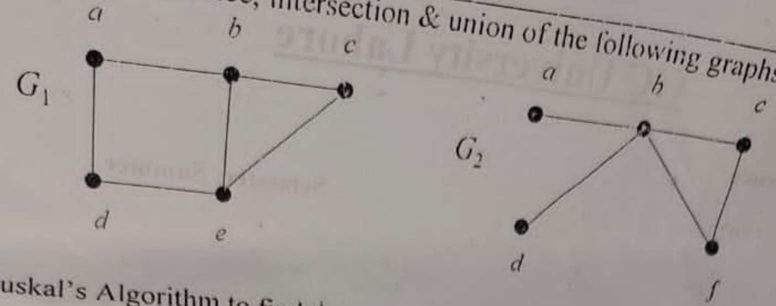
Roll No:

Subjective

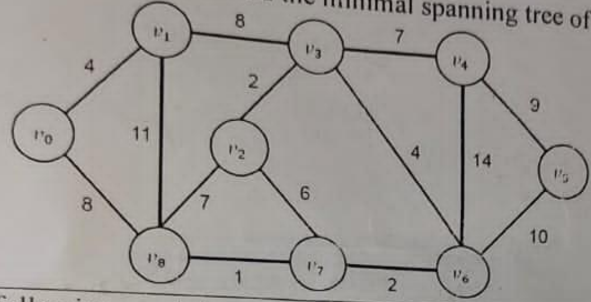
Attempt any Four Questions

Q#	Question	Marks
2	a) What is the difference between Eulerian Path and Hamiltonian Path. Explain with help of example. b) Define Maximal and Minimal matching of graph with the help of example. c) A tree has five vertices of degree 2, three vertices of degree 3 and four vertices of degree 4. How many vertices of degree 1 does it have?	4+4+2
3(i)	Find Adjacency Matrix and Incidence Matrix of the given directed weighted graph.  5 6 5-6+f=2 -1+3=2 2=2	6
(ii)	Let G be a connected planar graph with n vertices, m edges and f faces. Then prove that $n-m+f=2$.	4
4(i)	Define Edge coloring, Chromatic number and Four-Color theorem with help of example.	5
(ii)	Find chromatic polynomial of the given graph.  73 71 6 5 3	5
5(i)	Use Kruskal's Algorithm to find the minimal spanning tree of the following graph. 	5
(ii)	Draw all spanning tree of K_5 Graph.	5

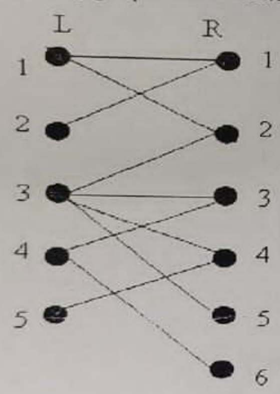
5(i)	Find symmetric difference, intersection & union of the following graphs.	5



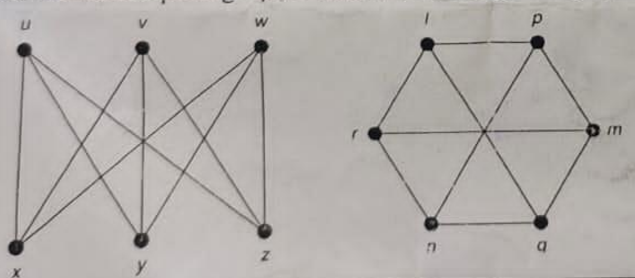
(ii)	Use Kruskal's Algorithm to find the minimal spanning tree of the following graph.	5



6(i)	Does the following graph have maximal matching?	5



(ii)	Define Isomorphic graphs. Show that the given graphs are isomorphic or not.	5



Department of Computer Sciences, GC University Lahore

BSCS FINAL-TERM EXAMINATION FALL-23

SUBJECT: **Graph Theory (CS-4102) (Subjective)** MAX MARKS: 40

SEMESTER: 5th TIME ALLOWED: **135 mnts**

Note: Solve any 4 questions.

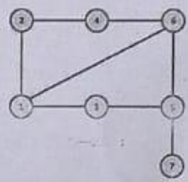
Q.2: **(6+4)**

a) Suppose that a group of ministers serve on committees as described below:

Committee	**Members**
Culture, Media & Sport	Ali, Babar, Kaleem
Defense	Kaleem, Danial, Imran
Education	Ali, Ghias
Food & Rural Affairs	Danial, Faheem
Foreign Affairs	Imran, Haroon
Justice	Babar, Imran, Ghias
Technology	Kaleem, Faheem, Haroon

What is the minimum number of time slots needed so that one can schedule meetings of these committees in such a way that the ministers involved have no clashes?

b) Find Radius and Diameter of the given graph.



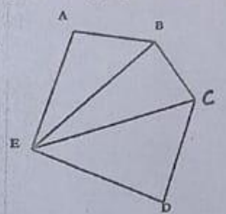
Q.3:

(6+4)

a) Draw an activity digraph from the given data and find critical path.

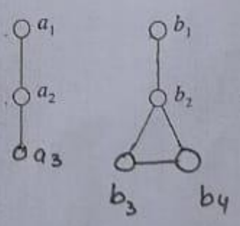
Nodes	Predecessors	Time Duration
A	-	7
B	-	2
C	-	15
D	A, B	10
E	D	8
F	D	5
G	E, F	2
H	F	8
I	C, G	2
J	I	3

b) Find out a vertex-disjoint subgraph and an edge-disjoint subgraph from the following graph.

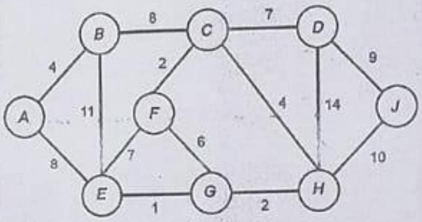


Q.4: (6+4)

b) Find the cartesian product of the following two graphs

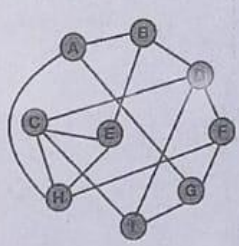


a) Write a suitable algorithm to find MST (Minimal Spanning Tree) and apply it to the given graph.

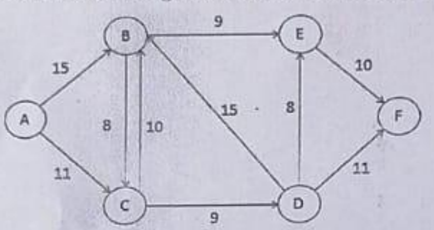


Q.5: (5+5)

a) Write Welsh Powell algorithm and then apply it to find optimal coloring of the given graph.

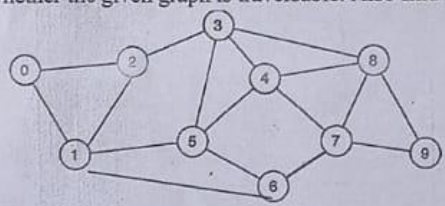


b) Apply Ford-Fulkerson Algorithm to find maximum flow for the given graph.

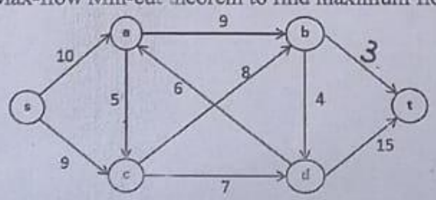


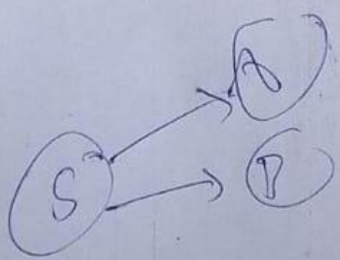
Q.6: (4+6)

a) Determine whether the given graph is traversable. Also find 2 Hamiltonian circuits if exist.



b) Apply Max-flow Min-cut theorem to find maximum flow of the given graph.





Gujranwala

Mid Term exams 2023

Department of computer science

Program: *BSCS V* **Shift:** **Section:**

Course Title: *Graph Theory* **Course Code** CS-4102

Instructor: *Ms. Anjum Ijaz* **Time Allowed:** 2 hr **Max. Marks** 50

INSTRCUTIONS:

1. Write **Name** and **Roll Number** on both Question Paper and Answer Book before start of Paper.
2. Attempt all Questions.
3. No. Extra Time will be given to student in any case.
4. No extra Sheet will be provided to the student.

ROLL NO.________ NAME:____________________

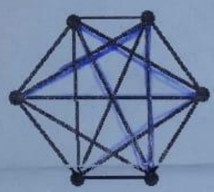
Subjective

Write the answer only four of the following questions.

Each question contains equal marks. (10 × 4 = 40)

1.

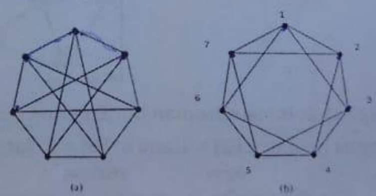
(i) **Check whether that the following graph is Hamiltonian or not.**



(ii) **Draw the graphs associated with the Pru¨fer code (5,1,3,6,2,7) and (5,4,5,1,3,9).**

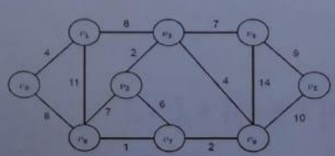
2. (5+5=10)

(i) **Check the Isomorphism of the following Graph.**

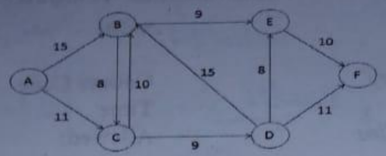


3.

(i) Draw a minimal spanning tree of the following Graph by Prims Algorithm. (5+5=10)

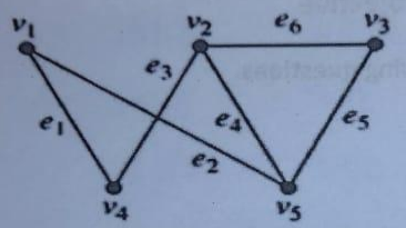


(ii) Use for-Fulkerson Algorithm to find the maximum flow for the following network.

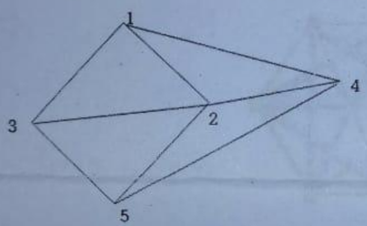


4.

(i) Find the incident and adjacent matrix of following **(5+5=10)**
Graph.

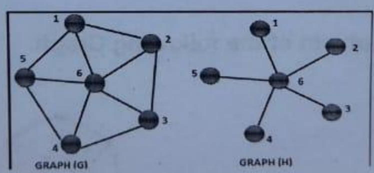


(ii) Find the chromatic polynomial of the given graph.



5. (3+3+4=10)

(i) Find symmetric difference intersection and union of the graphs.



(ii) Draw a graph that is Hamiltonian but Eulerion and draw a graph that is Eulerion but not Hamiltonian.

(iii) Draw a graphs having κ (G) = 1 and κ' (G) = 3. Also draw a graph with κ'(G) = 3 and δ(G) = 3

egde vertex vertex degree

4+6+7+9+2+4+1+2

10 + 10 + 8 + 7